



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SRI VELAVAA TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

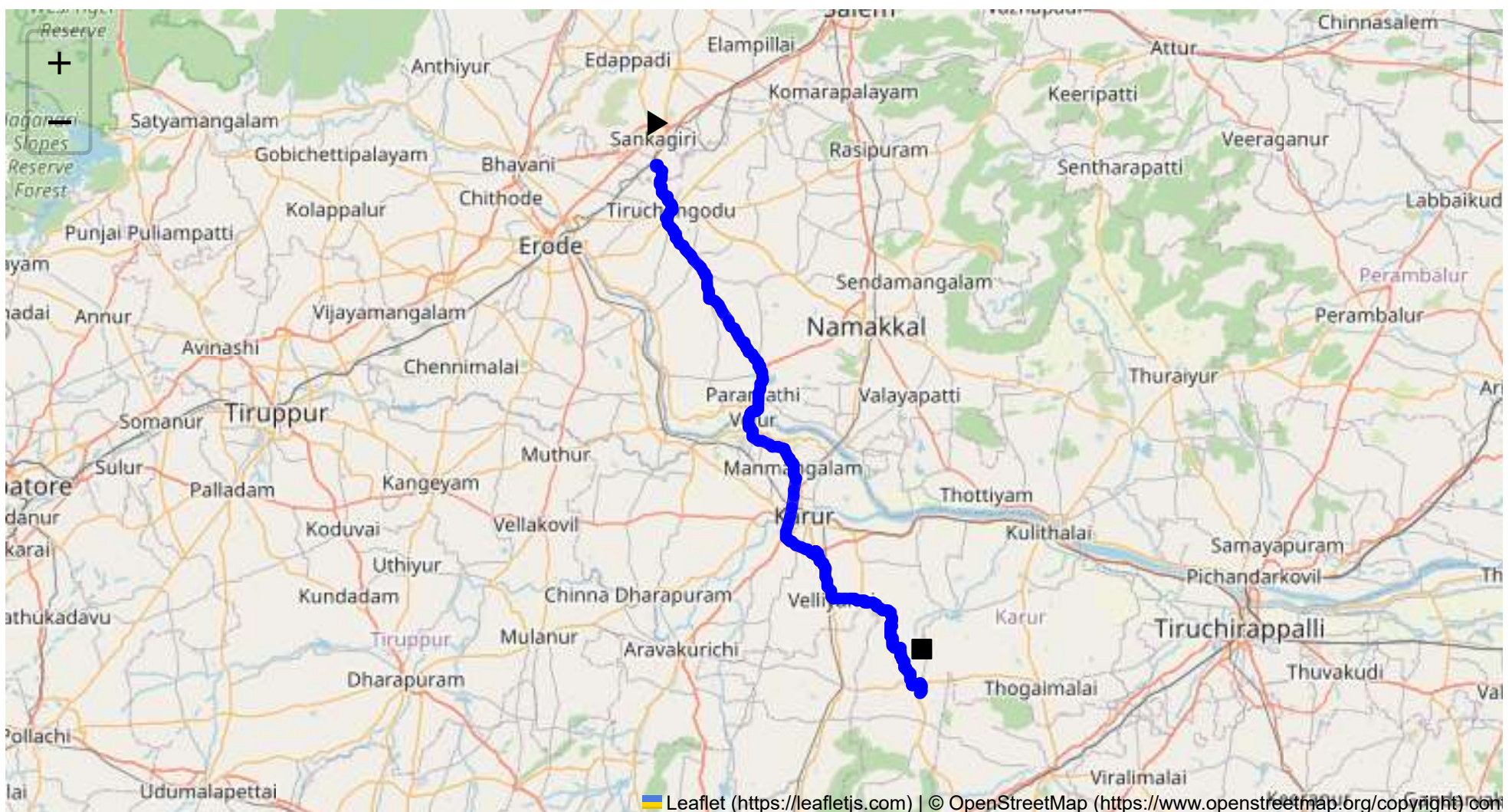
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 107.85 km
Estimated Duration: 2.2 hours
Adjusted Duration (Heavy Vehicle): 2.8 hours
Start: (11.4381, 77.8734)
End: (10.716029, 78.241745)



Welcome to the Journey Risk Management Study

Route Safety Analysis for CVQF+23W, Sangagiri to P68R+9PX, Keelappaguthi

1. Overview of the Route Map:

- The route from Sangagiri to Keelappaguthi covers approximately 108 kilometers via state highways and rural roads. It primarily traverses through agricultural regions with small towns and villages along the path.

2. Weather Conditions and Hazards:

- Tamil Nadu experiences a tropical climate with hot summers (March to June), monsoon rains (July to September), and relatively mild winters (November to February).
- During monsoon, heavy rains can cause waterlogging and poor visibility. Drivers must prepare for potential slippery roads and flooding, particularly in low-lying areas.

3. Traffic Patterns:

- Major traffic congestion occurs in the morning (7:30-9:30 AM) and evening (5:00-7:00 PM) due to local commuter traffic and transport activities.
- Look out for increased congestion near major junctions and market towns, such as Salem and Namakkal.

4. Road Quality and Infrastructure:

- The route includes both well-maintained highways and less developed rural roads. Some segments may have potholes, uneven surfaces, or narrow passages, particularly in rural areas.
- Roadworks or construction may occur periodically, potentially causing delays.

5. Alternative Routes:

- In case of an emergency or blockage, consider rerouting via NH44 or NH79, which are generally in better condition and offer quicker detours.

6. Local Regulations on Hazardous Material Transport:

- Hazardous materials transport is subject to the Motor Vehicles Act and specific state regulations. Restrictions often apply in urban areas during peak hours.
- Ensure compliance with safety signage, protective gear requirements, and emergency equipment mandates.

7. Historical Incidents Involving Heavy Vehicles:

- There have been incidents involving heavy vehicles overturning due to sharp turns in rural areas or during extreme weather conditions. Awareness and caution are advised, especially on unfamiliar roads.

8. Environmental Considerations:

- The route passes through agricultural and occasionally sensitive ecological zones, such as water bodies and protected lands.
- Minimize noise and emissions, and be cautious of wildlife crossings and agricultural machinery.

9. Communication Coverage:

- Major towns and highways generally have good cellular coverage. However, rural segments may experience dead zones, especially in hilly or dense vegetative areas. Pre-plan communication accordingly.

10. Emergency Response Times:

- Near urban centers like Salem or Namakkal, emergency response times are typically under 30 minutes. Rural areas may experience longer wait times. Familiarize yourself with the locations of nearby hospitals and emergency services.

11. Overall Summary of Risk Assessment:

- The route is generally safe if proper precautions are observed. Major risks include weather-related hazards during the monsoon, potential traffic congestion, and road quality issues in rural areas.
- Drivers should remain vigilant, especially in unfamiliar or less developed sections, and be prepared for sudden changes in road conditions or regulations affecting hazardous materials.

This assessment aims to equip drivers with the necessary knowledge to safely navigate their journey while maintaining observance of local law and ensuring environmental responsibility.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	U-Turn	High	10.9110477, 78.09713769999999	10 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44019, 77.87547	15 KM/Hr
5	Turn	Medium	11.39685, 77.88861	30 KM/Hr
6	Turn	High	11.37862, 77.89488	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16950, 78.01544	30 KM/Hr
9	Turn	Medium	11.16764, 78.01565	30 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Turn	High	11.14480, 78.02291	15 KM/Hr
12	Turn	Medium	11.10102, 78.00572	30 KM/Hr
13	Turn	Medium	11.05261, 78.05138	30 KM/Hr
14	Turn	Medium	10.90767, 78.09893	30 KM/Hr
15	Turn	High	10.90793, 78.09916	15 KM/Hr
16	Blind Spot	Blind Spot	10.91105, 78.09714	10 KM/Hr
17	Turn	High	10.84613, 78.12000	15 KM/Hr
18	Turn	Medium	10.84321, 78.16436	30 KM/Hr
19	Turn	High	10.84057, 78.16518	15 KM/Hr
20	Turn	High	10.83953, 78.18108	15 KM/Hr
21	Turn	Medium	10.83523, 78.18173	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
22	Turn	Medium	10.82892, 78.20166	30 KM/Hr
23	Turn	Medium	10.82865, 78.20188	30 KM/Hr
24	Turn	Medium	10.80081, 78.20092	30 KM/Hr
25	Turn	Medium	10.79274, 78.20214	30 KM/Hr
26	Turn	Medium	10.79217, 78.20276	30 KM/Hr
27	Turn	Medium	10.79013, 78.20307	30 KM/Hr
28	Turn	Medium	10.78767, 78.20438	30 KM/Hr
29	Turn	Medium	10.78380, 78.20278	30 KM/Hr
30	Turn	Medium	10.78037, 78.20649	30 KM/Hr
31	Turn	High	10.77911, 78.21461	15 KM/Hr
32	Turn	Medium	10.77330, 78.21431	30 KM/Hr
33	Turn	High	10.77239, 78.21467	15 KM/Hr
34	Turn	High	10.77215, 78.21560	15 KM/Hr
35	Turn	High	10.76401, 78.21632	15 KM/Hr
36	Turn	Medium	10.75296, 78.22112	30 KM/Hr
37	Turn	Medium	10.75178, 78.22246	30 KM/Hr
38	Turn	Medium	10.74981, 78.22308	30 KM/Hr
39	Turn	Medium	10.74551, 78.22709	30 KM/Hr
40	Turn	Medium	10.73084, 78.23146	30 KM/Hr
41	Turn	High	10.72910, 78.23136	15 KM/Hr
42	Turn	Medium	10.72872, 78.23407	30 KM/Hr
43	Turn	High	10.73159, 78.24282	15 KM/Hr
44	Turn	Medium	10.73136, 78.24316	30 KM/Hr
45	Turn	Medium	10.72890, 78.24432	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
18	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.92958, 78.05452



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 10.9110477, 78.09713769999999



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44019, 77.87547



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16950, 78.01544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14545, 78.02208



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.10102, 78.00572



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.05261, 78.05138



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.90767, 78.09893



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.90793, 78.09916



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 10.91105, 78.09714



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.84613, 78.12000



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.84321, 78.16436



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.84057, 78.16518



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.83953, 78.18108



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.83523, 78.18173



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.82892, 78.20166



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.82865, 78.20188



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.80081, 78.20092



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79274, 78.20214



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79217, 78.20276



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.79013, 78.20307



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.78767, 78.20438



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.78380, 78.20278



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.78037, 78.20649



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.77911, 78.21461



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.77330, 78.21431



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.77239, 78.21467



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.77215, 78.21560



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.76401, 78.21632



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.75296, 78.22112



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.75178, 78.22246



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.74981, 78.22308



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.74551, 78.22709



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.73084, 78.23146



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.72910, 78.23136



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.72872, 78.23407



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.73159, 78.24282



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.73136, 78.24316

